



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies



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A multi-cloud platform for secure, HIPAA-compliant de-identification and sharing of medical imaging data — for research teams and radiology departments alike

In Greek mythology, the aegis was the divine shield of Zeus and Athena — a symbol of protection. The name captures our mission: shielding patient identity while enabling the free flow of imaging data for research and clinical care.

Matthew L. Senjem, M.S.

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The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray — for multi-site clinical trials, research collaborations, and second opinions. Doing this safely is harder than it should be.

De-identification tools frequently fail. A 2015 study in *European Radiology* tested 10 free DICOM de-identification tools against 50 required PHI data elements. With default settings, only 1 of 10 tools successfully removed all required elements; 4 tools achieved success rates of 26% or less.^[1]

Metadata removal alone is not enough for brain imaging. A 2019 *New England Journal of Medicine* correspondence from Mayo Clinic showed that automated face-recognition software correctly matched de-identified brain MRI participants to their photographs in **83% of cases** — using only the scan geometry, no metadata at all.^[2] This means head imaging requires an additional step: automated defacing (removal of facial surface features from the 3D volume).

Burned-in PHI is invisible to tag-level tools. Patient names, dates of birth, and accession numbers are routinely overlaid directly on image pixels — particularly in ultrasound, CT topograms, and fluoroscopy. Standard DICOM de-identification tools operate on metadata only and miss this entirely.^[3]

Installing software at sending sites is a barrier. Platforms like XNAT require a Java-based desktop client or server daemon installed and IT-approved at every participating hospital. MIRC CTP requires a standalone Java application. Each new site adds months of IT negotiation.

Data sharing is now federally mandated — but tooling lags. The NIH Data Management and Sharing Policy (NOT-OD-21-013, effective January 25, 2023) requires all NIH-funded investigators to share scientific data as a condition of their grant award, no later than the time of first publication.^[4] Institutions are now obligated to share data they have no reliable, low-friction way to de-identify and transmit.

Result: Multi-site studies are delayed by months. In a 2021 study of global randomized trials, contract execution alone averaged 7.9 months per US site.^[5] Software installation and IT approval add to this.

Market Opportunity

The medical image exchange market is growing, driven by federal data sharing mandates and expanding multi-site clinical trial volume in oncology and neurology.

Market	2024 Estimate	2034 Projection	CAGR
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Medical Image Exchange Systems [6]	\$3.9B	\$9.9B	9.7%
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Why now:

- The NIH DMS Policy (2023) creates a compliance obligation for tens of thousands of active grants that did not previously require a data sharing plan
- A 2024 NCI-sponsored benchmark (MIDI-B, presented at MICCAI 2024) formally tested de-identification tools and found that burned-in pixel PHI and free-text fields remain the hardest unsolved problems — no existing tool handles all cases reliably[\[7\]](#).
- Growing multi-site trial volume in oncology (PSMA PET, amyloid PET, whole-body MRI) and neurology (tau PET, fMRI) is driving demand for scalable data collection from distributed hospital networks
- Cloud-hosted DICOM storage (GCS, S3, Azure Blob) and managed databases (Cloud SQL, RDS, Azure Database) are now mature commodity infrastructure, reducing the barrier to building managed platforms

The Solution

AEGIS is a multi-cloud platform that makes secure medical image sharing as easy as uploading a file — for any DICOM modality. It runs on GCP, AWS, or Azure — or locally via Docker Compose. It serves two complementary audiences from a single platform.

Capability	What It Does
No software to install	Upload portal runs entirely in the web browser. Hospitals use their existing computers and internet connection — no desktop app, no IT approval for new software.
Tag-level anonymization in the browser	Patient data is stripped from DICOM tags before it leaves the hospital network, following the DICOM PS3.15 Annex E Basic Confidentiality Profile — the established international standard covering all 18 HIPAA Safe Harbor identifier categories. [8]
Automated defacing for head scans	MRI, CT, and PET studies of the head and brain undergo server-side defacing (removal of facial surface geometry) before entering the research archive. Non-head studies bypass this step automatically.
Burned-in PHI detection	Automated OCR scans image pixels for overlaid text (patient names, dates, accession numbers) that tag-level tools miss. Studies with detected pixel PHI

	are flagged for review.
Supports all DICOM modalities	MRI, CT, PET, PET/CT, ultrasound, X-ray, mammography, nuclear medicine, and more — using the same upload and anonymization workflow.
MRI protocol compliance	Automated verification that acquisition parameters (TR, TE, flip angle, resolution) match site-specific templates per scanner manufacturer, model, and software version — catching the 0.19–64% non-compliance rates found across multi-site studies. [10]
Centralized audit trail	Every upload, approval, rejection, and data export is logged with a timestamp and actor. Institutions can demonstrate HIPAA compliance from a single dashboard.
Clinician review before release	Administrators review anonymized images in a web-based DICOM viewer (OHIF) before approving studies for sharing. Nothing is shared automatically without human sign-off.

Who It's For

AEGIS serves two audiences from a single de-identification and routing platform:

Multi-site research teams need to collect imaging data from hospitals they don't control, de-identify it to meet NIH and IRB requirements, and convert it to research-ready formats. AEGIS gives them a browser link they can send to any participating site — no software installation, no IT negotiation. The research pipeline (protocol compliance, automated QC, BIDS conversion, defacing) ensures data arrives analysis-ready.

Hospital radiology departments need to de-identify imaging data for internal uses — training AI models, building real-world evidence databases, sharing with referring physicians, and fulfilling data requests from payers and registries. AEGIS gives them a centralized de-identification gateway with a complete audit trail, burned-in PHI detection, and configurable routing rules — without replacing their PACS or requiring a separate enterprise integration project.

	Research Teams	Radiology Departments
Entry point	Browser upload link sent to external sites	DIMSE receive / internal batch import
Key need	NIH DMS compliance, multi-site consistency	HIPAA de-identification, audit trail

Differentiator	Zero-install at sending sites	No PACS/VNA integration required
Pipeline features	Defacing, protocol compliance, QC, BIDS	Burned-in PHI, routing rules, export shares
Review workflow	PI/coordinator approves before archive	Radiologist/admin approves before release

The underlying platform — de-identification engine, routing rules, audit trail, OHIF viewer, and admin dashboard — is shared. Research-specific features (defacing, BIDS, protocol templates) and enterprise features (DIMSE receive, HL7/FHIR notifications, tag standardization) extend it for each audience.

How De-identification Works

AEGIS follows a two-phase approach that addresses the limitations documented in the literature:

- Phase 1 – In the browser (before upload):
- └─ DICOM tags stripped per PS3.15 Annex E Basic Profile [\[8\]](#)
 - └─ 18 HIPAA Safe Harbor identifiers addressed [\[9\]](#)
 - └─ Reviewer sees before/after tag comparison before confirming
 - └─ Only de-identified data is transmitted – PHI never leaves the hospital
- Phase 2 – On the server (after upload):
- └─ De-identification completeness check
 - └─ OCR scan of image pixels for burned-in text [\[3\]](#)
 - └─ Head imaging → automated defacing [\[2\]](#)
 - └─ Administrator review and approval before data is shared

This two-phase design directly addresses the gaps identified in the Aryanto (2015) [\[1\]](#) and Schwarz (2019) [\[2\]](#) studies.

Supported Modalities

Category	Modalities
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Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images

Initial focus: Brain MRI, PET, and CT — these exercise the full pipeline including defacing. All other modalities use the same upload and de-identification workflow.

Technology

Layer	Technology	Notes
Backend	Go on Cloud Run (GCP) or ECS Fargate (AWS)	Compiled binary, minimal dependencies, fast cold starts
Frontend	React + TypeScript	Runs in any modern browser, no installation required
Database	PostgreSQL 15 (Cloud SQL / RDS)	Audit trail, project/user management, routing rules
DICOM Storage	Cloud-neutral (GCS, S3, or local filesystem)	Abstracted behind a pluggable storage interface
OCR / PHI Detection	Tesseract OCR (local) / cloud AI (pluggable)	Detects burned-in text in image pixels
Defacing	DeepDefacer (default), mri_deface, mri_reface	Multiple backends with automatic fallback; see docs/research/mri-defacing-tools-comparison.md
DICOM Viewer	OHIF Viewer (v3)	Open-source, browser-based, supports all modalities
Auth	GCP IAP / AWS ALB+Cognito / Azure AD	Multi-provider auth middleware, auto-detection
Processing Pipeline	7 Python services (6 processing + 1 DIMSE receiver adapter)	Classification, PHI detection, protocol compliance, QC, defacing, BIDS conversion, plus DIMSE C-STORE ingress — auto-dispatched in dependency order

Infrastructure	Terraform (GCP + AWS modules), Docker Compose	Reproducible, version-controlled, multi-cloud; local dev stack starts everything with one command
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On the use of automated tools: AEGIS uses automated tools to assist with — not replace — human review. Automated de-identification flags potential issues; a trained administrator reviews and approves every study before it is shared. Automated defacing quality is reviewed side-by-side against the original in the admin interface.

Compliance Framework

AEGIS is designed around established regulatory and technical standards:

Standard	Reference	What It Covers
HIPAA Privacy Rule	45 CFR § 164.514(b)	De-identification methods: Safe Harbor (18 identifiers) and Expert Determination
DICOM Confidentiality Profile	NEMA PS3.15 Annex E	Tag-level de-identification actions for all DICOM attributes
NIH Data Management Policy	NOT-OD-21-013 (eff. Jan 25, 2023)	Data sharing obligations for NIH-funded investigators
Cloud Security	BAA, VPC/VPC-SC, CMEK/KMS	Business Associate Agreement, encrypted storage, network controls — available on GCP, AWS, and Azure

AEGIS does not create a new de-identification standard — it implements the ones that already exist.

Competitive Landscape

	AEGIS	ENCOG (Enlitic)	XNAT	Flywheel	MIRC CTP
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Hosting	Multi-cloud (GCP, AWS, Azure)	On-premises / hybrid	Self-hosted	Commercial SaaS	On-premises
Install at sending site	None — browser only	PACS/VNA integration	Desktop Java client	CLI tool	Java application
Burned-in PHI	OCR detection	AI CV detection	None	Built-in	None
Defacing	Automated, reviewable	None	Manual or plugin	Built-in	None
Modalities	All DICOM	MR, CT, XR, US	Neuroimaging focus	All imaging	All imaging
Open source	Private (built on open-source stack)	No (commercial)	Yes	No	Yes
Audit trail	Centralized, per-study	Chain of custody	Per-instance	Built-in	Limited

Existing platforms tend to serve either research (XNAT, Flywheel) or enterprise radiology (ENCOG/Enlitic, MIRC CTP) — but not both. AEGIS is designed for both audiences from a shared platform, and is the only option that runs on any major cloud provider without re-architecture.

Enlitic's ENCOG is the closest commercial analogue to AEGIS's de-identification pipeline.^[12] It uses AI-driven computer vision to detect burned-in text overlays and claims to protect over 4,000 DICOM fields. However, ENCOG requires PACS/VNA integration at each sending site, does not offer volumetric defacing for head imaging, and publicly documents only four modalities (MR, CT, XR, ultrasound). ENCOG is part of a broader commercial suite (Ensign) that includes data standardization (ENDEX, FDA 510(k) cleared) and migration tools — but lacks the research pipeline features (protocol compliance, QC, BIDS) that multi-site studies require.

XNAT and Flywheel serve research well but require software installation at sending sites and offer limited enterprise radiology integration. MIRC CTP is a robust open-source pipeline but requires on-premises Java infrastructure and has no burned-in PHI detection or defacing.

Key differences from existing platforms:

1. **No software installation at sending sites** — the full anonymization workflow runs in the browser. ENCOG, XNAT, and MIRC CTP all require software deployed at the sending institution.
2. **Two-phase de-identification** — both tag-level (browser) and pixel-level (server-side OCR) are addressed in a unified pipeline

3. **Defacing is automated and reviewable** — not optional, manual, or absent. Neither ENCOG nor MIRC CTP offer volumetric defacing.
 4. **Not specialty-specific** — designed for all DICOM modalities from the start, unlike XNAT (neuroimaging focus) or ENCOG (4 listed modalities)
 5. **Dual-market platform** — research features (protocol compliance, QC, BIDS) and enterprise features (DIMSE receive, routing rules, tag standardization) extend a shared de-identification core. Competitors serve one audience or the other.
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Phased Roadmap

*Development status note: Phases 1–4 are **code-complete** — all features are implemented, compile, and pass CI. Recent hardening shipped includes timezone-stable rendering controls across UIs and strict importer contract validation (canonical selectors, strict JSON, absolute-path requirement). The platform has not yet been deployed to a cloud environment or tested with real clinical data. A test deployment on GCP or AWS is planned as the next milestone.*

Phase 1 — Foundation (Code Complete)

- Cloud infrastructure (Terraform for GCP and AWS)
- Browser-based upload portal with DICOM tag anonymization and before/after preview
- Go API with DICOM ingest and PostgreSQL audit trail
- Admin dashboard with OHIF viewer for QC, approve/reject workflow

Phase 2 — Defacing + Notifications (Code Complete)

- Automated mri_deface pipeline for head imaging
- Side-by-side before/after defacing review in admin interface
- Email notifications for upload confirmation, approval, and rejection

Phase 3 — Operations (Code Complete)

- Multi-project routing engine with configurable rules by modality, source, and anatomy
- Institution management with project-scoped roles
- Configurable anonymization profiles per project (which tags to retain for research)

- Audit log viewer with filtering
- Admin user management, RBAC enforcement (admin/viewer roles), authentication middleware

Phase 4 — Advanced Processing (Code Complete)

- Burned-in PHI detection using OCR on image pixels (addresses gap identified in [\[3\]](#), [\[7\]](#))
- Automated image quality assessment (motion artifact detection, coverage completeness)
- MRI protocol compliance — per-scanner, per-sequence parameter validation against configurable templates with tolerances, following the approach used by ADNI^[11] and tools like mrQA^[10]
- BIDS format conversion for neuroimaging research output
- Batch import tools for historical data migration
- Multi-cloud support — AWS S3 storage backend, ALB + Cognito auth, Terraform AWS module

Next Milestone — Test Deployment

- Deploy to GCP or AWS using Terraform modules
- End-to-end testing with sample DICOM datasets (TCIA public data)
- Validate full pipeline: upload → de-identify → deface → QC → approve → export
- Performance benchmarking and cost validation

Phase 5 — Enterprise Radiology

- DIMSE service-class expansion — add C-FIND/C-MOVE workflows beyond current C-STORE ingest
- DICOM tag standardization — normalize study/series descriptions across sites and scanners for consistent downstream processing
- HL7 FHIR notifications — notify hospital EMR/RIS systems when studies are de-identified, approved, or exported
- Reversible de-identification — organization-owned keys for internal use cases where re-identification may be required
- PACS/VNA query-retrieve — pull studies from hospital archives on demand rather than waiting for push
- Multi-tenant SaaS deployment — shared infrastructure with per-organization data isolation for radiology groups managing multiple facilities

Cost Estimate (Development / Proof of Concept)

Resource	GCP Monthly	AWS Monthly
Containers (Cloud Run / ECS Fargate)	~\$5–15	~\$5–15
PostgreSQL (Cloud SQL / RDS)	~\$10	~\$15
Object Storage (GCS / S3)	~\$0.02/GB	~\$0.02/GB
Load Balancer	Included	~\$16 (ALB)
Total (development)	~\$20–40/month	~\$40–60/month

Production costs scale with data volume. A 1,000-session multi-site study (~500 GB) would cost approximately \$50–100/month in storage on either cloud.

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AEGIS

Secure, browser-based medical image de-identification and sharing for any DICOM modality — serving both multi-site research teams and hospital radiology departments from a single multi-cloud platform, with automated defacing, pixel-level PHI detection, and administrator review before any data is released. Runs on GCP, AWS, or Azure.